# IT2020 – Software Engineering
## Year 2, Semester 1, 2024
## Group Assignment

Online Tour Guide

**Group Details**

**Campus: Malabe**

**SE/OOP Group Number: SE/OOP/2024/S2/MLB/WD/G70**

| | Student Registration Number | Student Name |
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## Introduction

“GuideMeNow” is an online platform designed to provide an interactive and user-friendly platform for travelers to explore and plan their trips. By connecting different roles: users, administrators, tour guides, drivers and bank, the system ensures that all stages of travel planning, booking and execution are well-coordinated. Each actor plays a specific role, contributing to the system's overall efficiency and user satisfaction.

When a user books a vehicle, they have to choose their interested places. After selecting the place, they can choose vehicles, a tour guide and other resources for the trip. Users can make their payments for tour guide, vehicle and other resources using a credit card, debit card and cash.

Registered users can login to the system using their login credentials and it is validated by the system. When a user registers, they can gain access to search and book tours. They can review and provide feedback about the service. System administrators can manage profiles and manage tours, manage payment and maintain the website and can provide support to customers to resolve any issue, ensuring quick response times for inquiries.

The tour guide is a crucial part of the user experience, responsible for conducting tours and ensuring that travelers have an enjoyable and informative journey. They can view and manage bookings assigned to them, ensuring proper coordination. They interact with users to clarify details, address concerns and provide personalized assistance before, during and after the tour and they lead the tour ensuring users a smooth tour experience.

Drivers can confirm transport requests initiated by users or tour guides, ensuring availability of vehicles and they can also edit their profiles. Drivers can communicate with the tour guides to confirm the travel information. The bank is also an important role when we consider payments. This handles the transfer of funds when a user books a tour or service. If there are cancellations or issues with bookings, the bank processes refunds to users efficiently.

These interconnected roles play a vital part in the system and effectively plan and execute tours. Users can browse and book tours, make secure payments through the bank system. They also receive real-time updates from their assigned tour guides. Drivers and tour guides are managed efficiently, ensuring timely and quality service. The admin is responsible for the entire system, managing user accounts, processing payments. This system gives a better experience with ensuring convenience, reliability and efficiency for their users.

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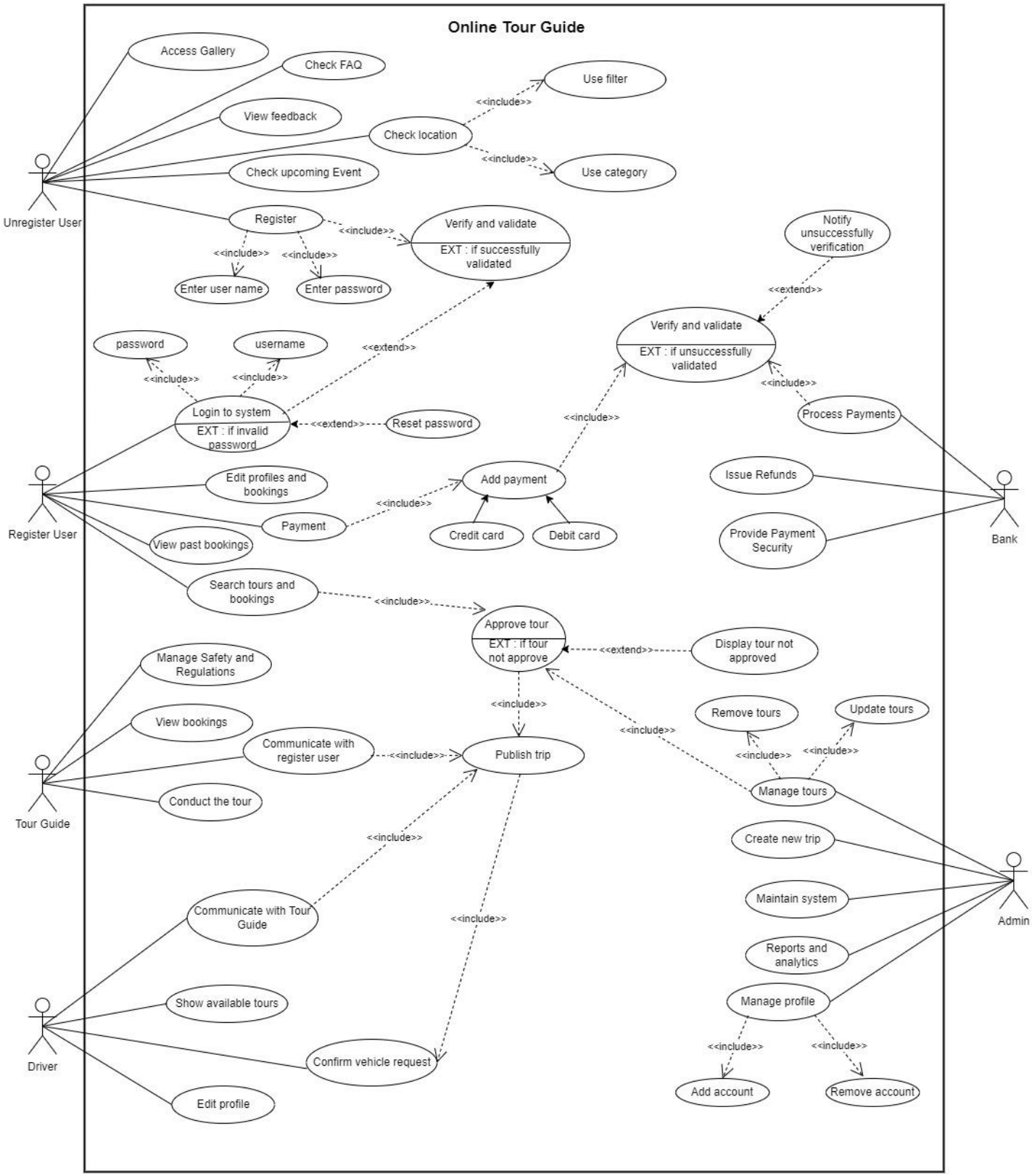
## Use Case Diagram



# IT23256750| W.M.V. Wijethunge

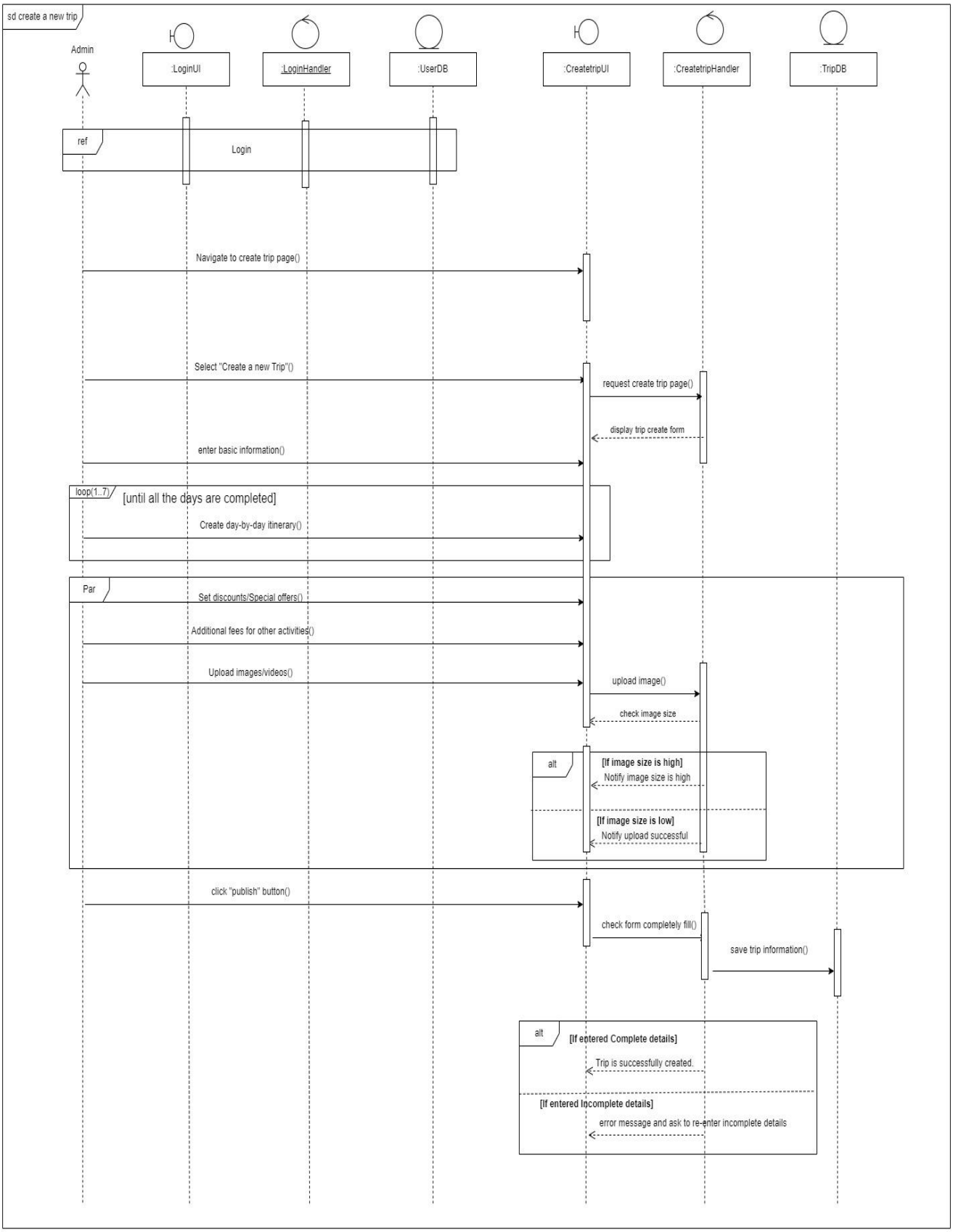
## Administrator creates a new trip.

### Use Case Scenario

| | | |
|---|---|---|
| Name | Create a new trip. | |
| Summary | This trip creation process enables administrators to customize travel plans based on user preferences and includes essential activities in it. | |
| Priority | 2 | |
| Pre-condition | The administrator must be logged into the system. | |
| Post-condition | A new tour is successfully created and published. | |
| Primary actor | Administrator | |
| Trigger | The registered user requests to create a new trip. | |
| Main scenario | Step | Action |
| | 1 | The system notifies the administrator of a new trip request submitted by the user. |
| | 2 | Administrator navigates to the login page. |
| | 3 | Enter credentials and login to the system. |
| | 3 | Administrator navigates to the trip management section. |
| | 4 | Administrator selects the option to “Create a new Trip”. |
| | 5 | Administrator enters basic trip information about the trip. |
| | 6 | The administrator creates the day-by-day itinerary. |
| | 7 | The administrator sets the discounts, and special offers . |
| | 8 | The administrator checks whether the optional activities are given by the user. |
| | 9 | The administrator uploads images, videos to the trip creation. |
| | 10 | After filling out all trip details, the administrator reviews the trip information. |
| | 11 | Then click the “publish” button to make the trip available on the website. |
| | 12 | Administrator receives a notification or email about successful trip creation. |
| Extensions | Step | Branching Action |
| | 1a | The request is denied if a trip cannot be accommodated. |
| | 1b | The admin informs the user and suggests alternatives. |
| | 3a | System displays an error message if the username or password is incorrect. |
| | 3b | System asks to re-enter username or password. |
| | 8a | If optional activities are given, set them into the trip creation. |
| | 9a | The system notifies the image size is high. |
| | 11a | The admin receives an error message. |
| | 11b | Ask to re-enter the incomplete details. |
| | 12a | Administrator does not receive any confirmation message or email about successful creation. |

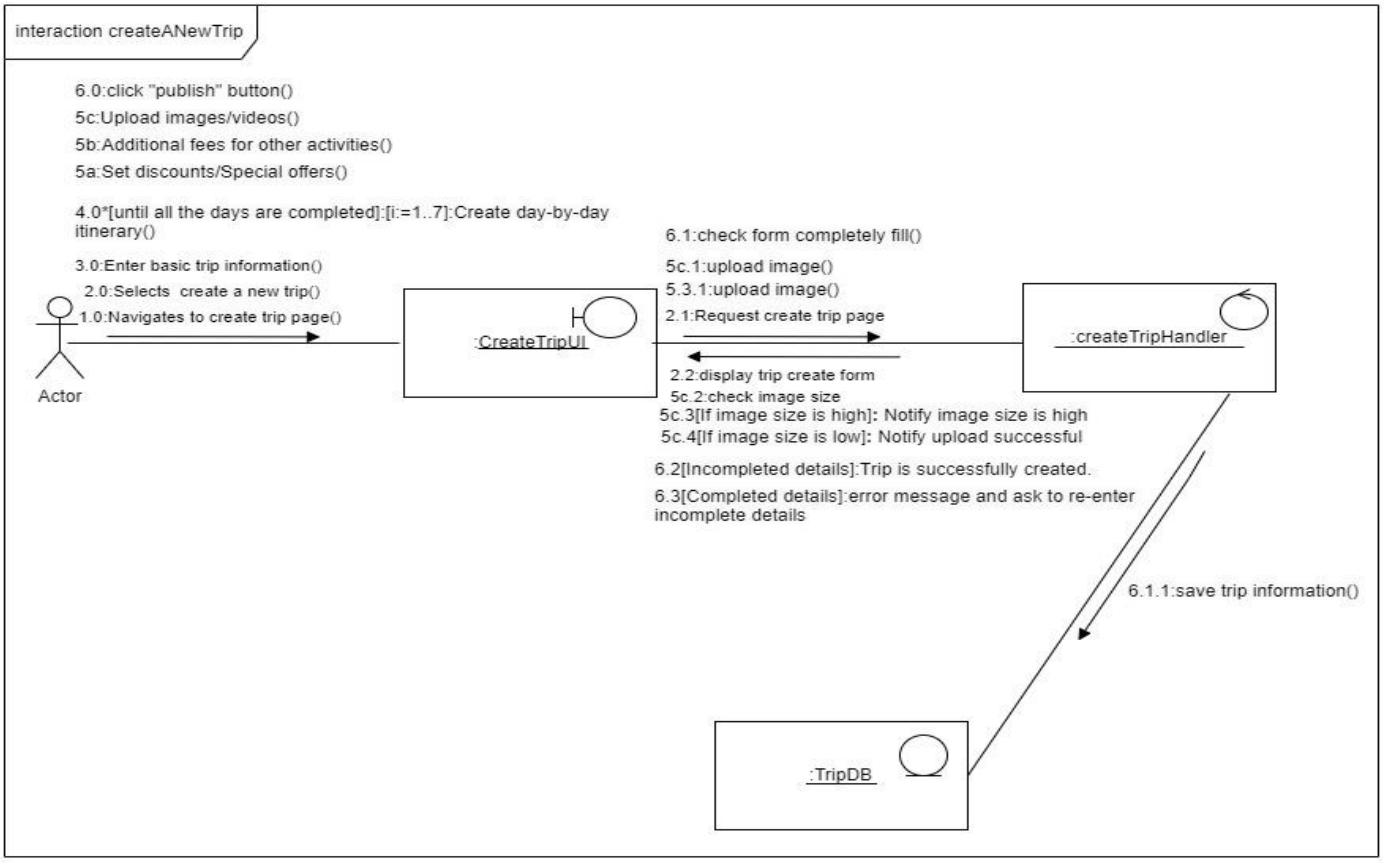
## Sequence Diagrams

**Create a new trip.**




## Communication Diagrams

**Create a new trip**



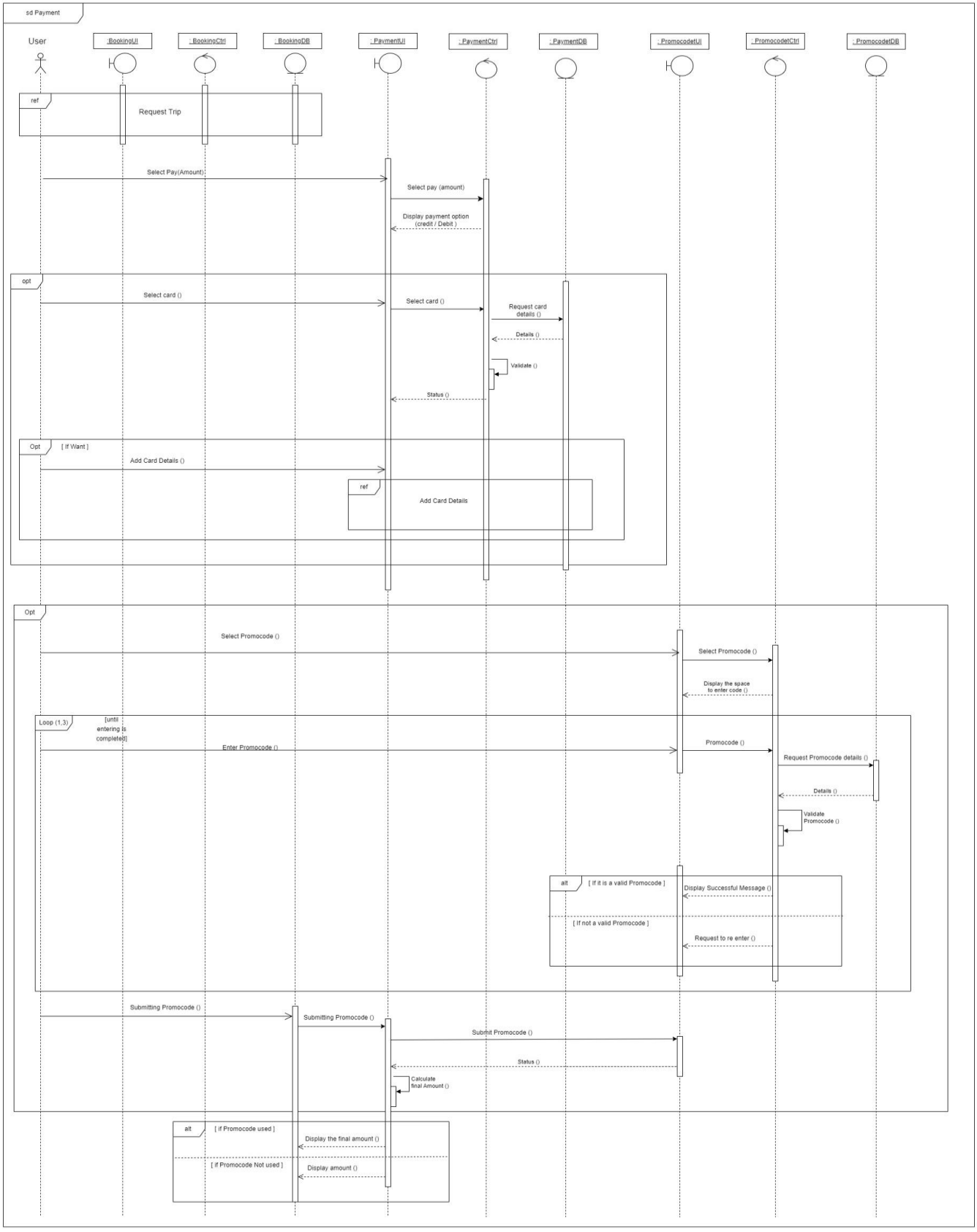
## IT23252936| W.L.S.A. Wijesingha

### Payment Process
### Use Case Scenario

| | | |
|---|---|---|
| **Name** | Payment process | |
| **Summary** | Pay for the trip | |
| **Priority** | 4 | |
| **Pre-condition** | The user must have selected a tour or trip package to pay for. | |
| **Post-condition** | Payment completed | |
| **Primary actor** | Registered user | |
| **Trigger** | To complete the booking process | |
| **Main scenario** | **Step** | **Action** |
| | **1** | Trip was confirmed by the passenger. |
| | **2** | Display the amount to be paid for the ride. |
| | **3** | Select the 'Pay' button and it will direct you to the Payment page. |
| | **4** | Select a payment method from credit or debit card payment. |
| | **5** | If card payment was selected, allow the user to select the saved card. |
| | **6** | System validates card details. |
| | **7** | When passengers use the app frequently, System sends emails containing promo codes . |
| | **8** | If a passenger needs to use the promo code, they can enter the code by selecting 'promo code' as an optional action. |
| | **9** | System will allow users to add the promocode in the provided space. |
| | **10** | Press 'Continue' the system validates the promo code. |
| | **11** | The system calculates the final value by reducing the discount. |
| | **12** | Return to the booking pages to continue further. |
| **Extensions** | **Step** | **Branching Action** |
| | **5a** | If the user wants to add a new card, select 'add payment method' and fill details |
| | **9a** | If the entered promo code is incorrect, show invalid message. |
| | **9b** | Allow passenger to re-enter it (3 times) |

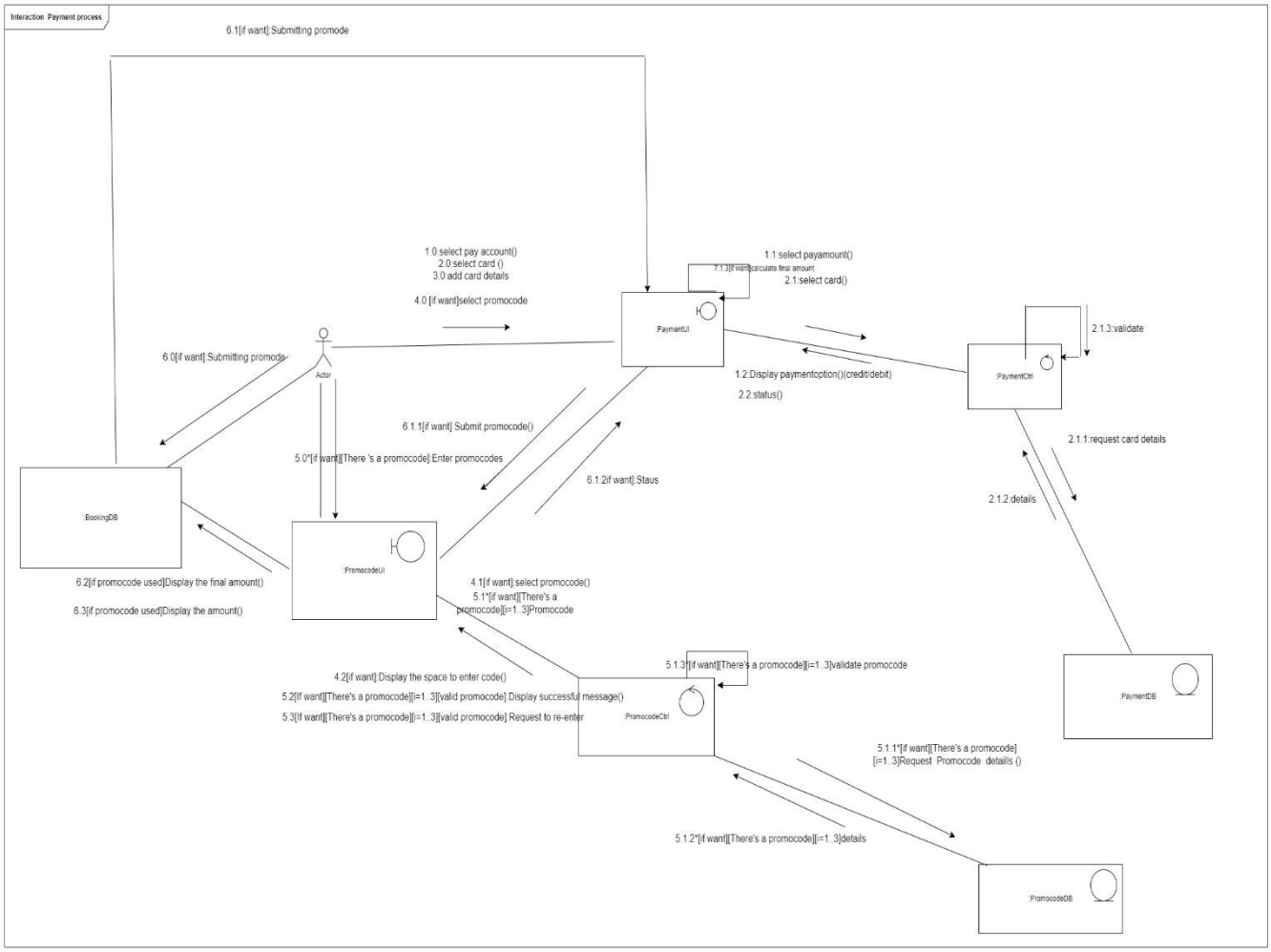
## Sequence Diagrams

Payment process



## Communication Diagrams

Payment process



# IT23243026| R.M.U.K. Rathnayake

## Driver update profile

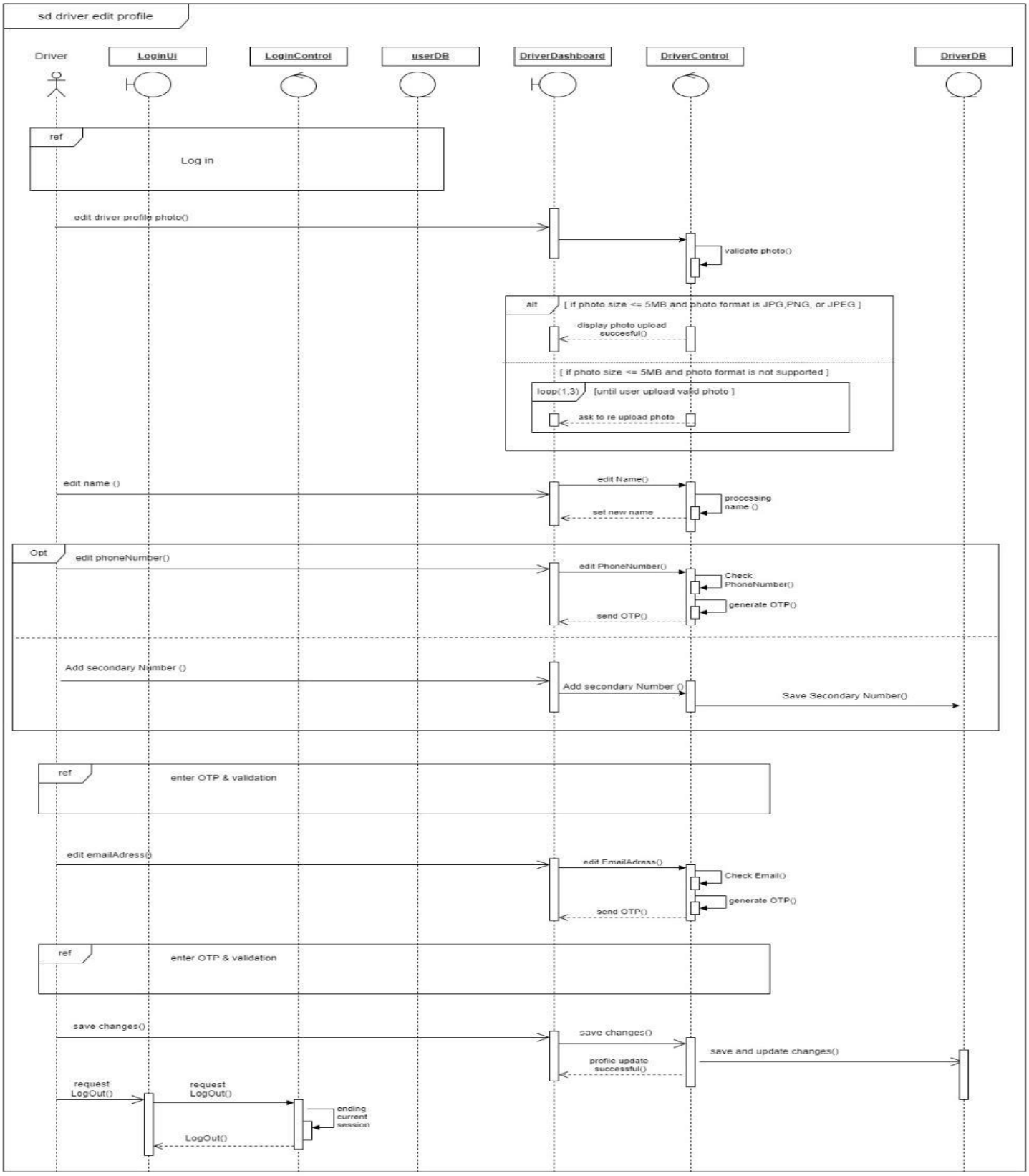
### Use Case Scenario

| | | |
|---|---|---|
| Name | Driver edits profile | |
| Summary | Driver modifies driver (user) profile and vehicle profile. | |
| Priority | 4 | |
| Pre-condition | The driver (user) needs to log into the system. | |
| Post-condition | The driver can change his profile. | |
| Primary actor | Driver | |
| Trigger | Driver select edit user profile. | |
| Main scenario | Step | Action |
| | 1 | Driver login to the system using his login credentials. |
| | 2 | Driver navigates to the Driver UI. |
| | 3 | The driver selects “Driver Profile”. |
| | 4 | Click on “Edit driver.”. |
| | 5 | The user clicks on “Edit profile photo.”. |
| | 6 | The system displays a window for selecting a photo to upload as a profile photo. |
| | 7 | The system displays “photo uploaded”. |
| | 8 | The user clicks on the First name and changes it. |
| | 9 | The user clicks on the Last Name and changes it. |
| | 10 | To add a phone number, the user clicks on it, enters the country code, then click "Add". |
| | 11 | The user inputs the verification code supplied to their cell phone. |
| | 12 | The system displays “Phone Number updated”. |
| | 13 | The user selects Email, inputs a new email address, and clicks "set". |
| | 14 | The user inputs the verification code delivered to their email address. |
| | 15 | System display “Email Updated”. |
| | 16 | The user clicks on “Save Changes”. |
| | 17 | System display “Updated Successfully”. |
| | 18 | The user logs out of the system. |
| Extensions | Step | Branching Action |
| | 1a | The system validates the entered login credentials. |
| | 1b | If the login credentials are invalid, the system will display an error notice and prompt you to re-enter them (up to 5 times, including this try). |
| | 1c | If the login credentials are invalid, the system will display an error notice and prompt you to re-enter them (up to 5 times, including this try). |
| | 6a | The system checks if the photo size is less than or equal to 5MB and the format is JPG, PNG, or JPEG. |
| | 6b | If not, the system will show "Invalid photo format or photo size exceeds 5MB." |
| | 10a | The system validates that the phone number format is accurate. |
| | 10b | If necessary, another phone number can be added |
| | 10c | The system sends a verification code to the provided cell phone number. |

| | | |
|---|---|---|
| | **11a** | **The system checks whether the code is valid or invalid.** |
| | **11b** | **If not valid, ask to re-enter the code.** |
| | **13a** | **The system determines whether the email address is legitimate.** |
| | **13b** | **If not, ask to re-enter a valid email address.** |
| | **13c** | **The system sends a verification code to the provided email address.** |
| | **14a** | **The system checks whether entered code is valid or invalid.** |
| | **14b** | **If not, ask to re-enter the code.** |

## Sequence Diagrams

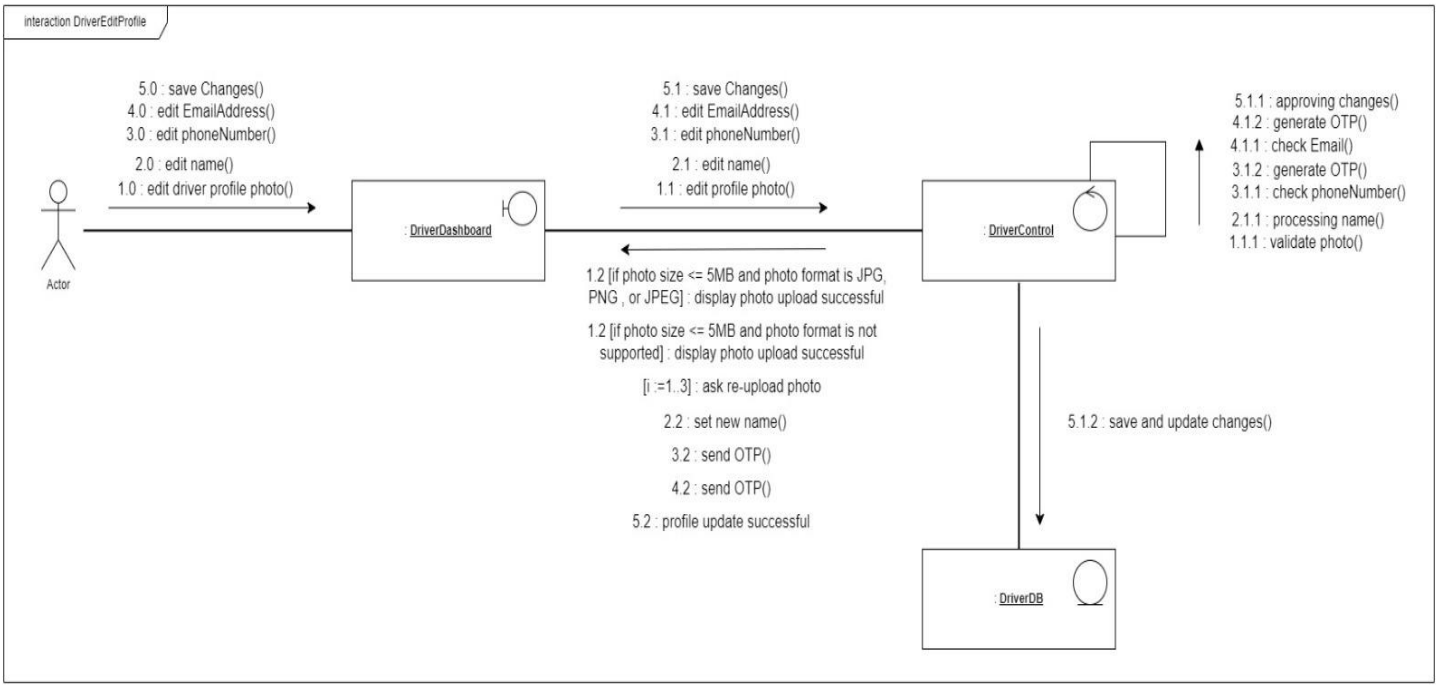
## Update Profile




**Sequence Diagram**

## Communication Diagrams

## Update Profile




**Communication Diagram**

# IT23150966| K.Y.T. Chamuditha

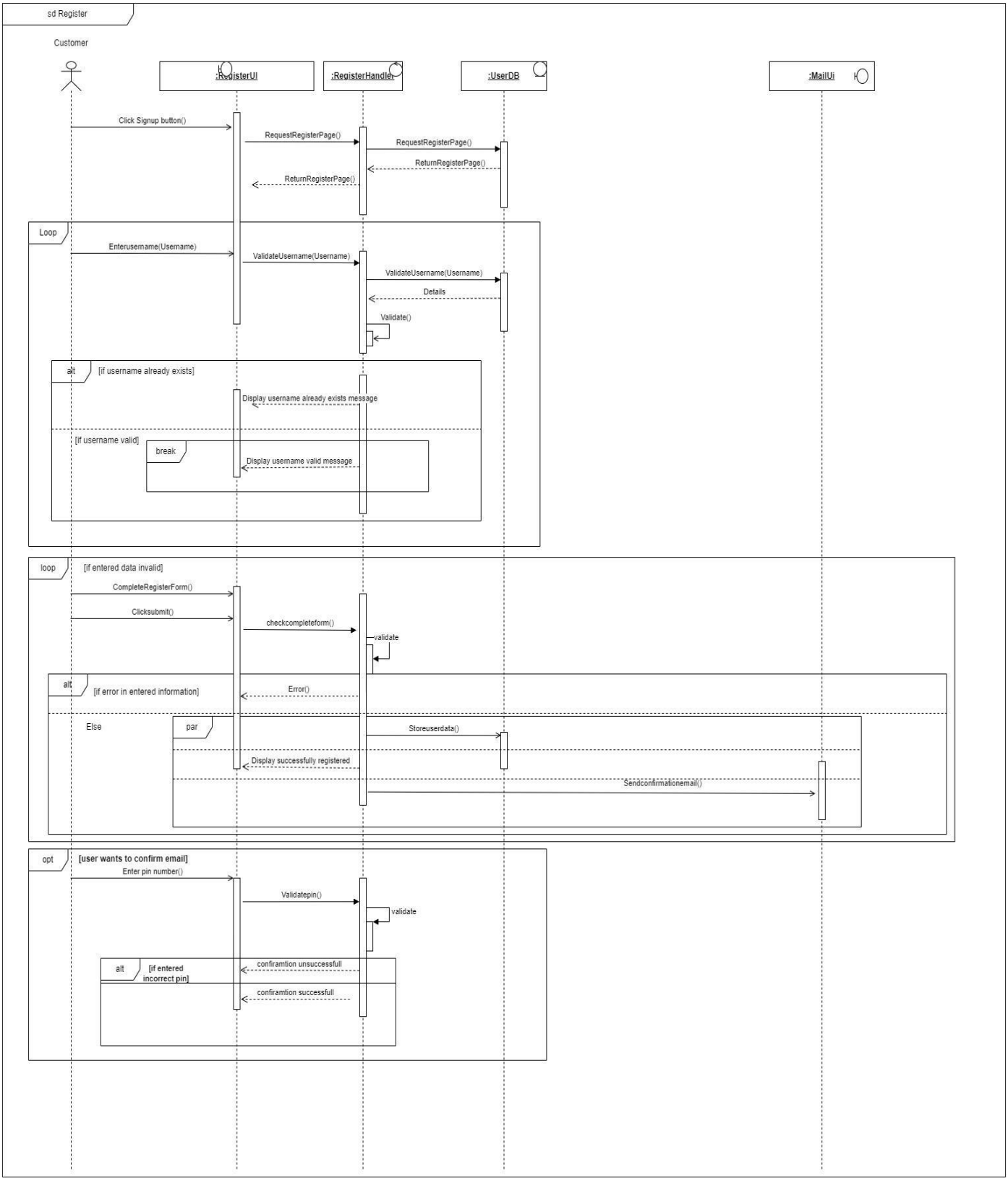
## User Login

## Use Case Scenario

| | | |
|---|---|---|
| **Name** | **Register** | |
| **Summary** | **Register as a user.** | |
| **Priority** | **01** | |
| **Pre-condition** | **Guests must go to the website.** | |
| **Post-condition** | **The system sends the user an email of appreciation, along with a message confirming successful registration.** | |
| **Primary actor** | **Guest User** | |
| **Trigger** | **Guest user selects to register.** | |
| **Main scenario** | **Step** | **Action** |
| | **1** | **Go to the website.** |
| | **2** | **Guest user click signup button.** |
| | **3** | **The system shows the registration form and prompts the user to input their name.** |
| | **4** | **The guest enters the login, full name, email address, and password using the specified characters and retypes the password.** |
| | **5** | **Guest clicks on the sign-up button to submit the registration form.** |
| | **6** | **The system saves the visitor information and sends an email with a PIN to validate the entered email.** |
| | **7** | **Guest user enter PIN on the website.** |
| | **8** | **The system sends the user a thank-you email, along with a confirmation of successful registration.** |
| **Extensions** | **Step** | **Branching Action** |
| | **5a** | **The system informs the guest that the entered username already exists.** |
| | **5b** | **The system alerts the guest that the supplied email address is not valid.** |
| | **5d** | **The system indicates to the guest that the re-entered password does not match the previous one.** |
| | **5e** | **The system notifies guests that the information given is incomplete.** |

## Sequence Diagrams

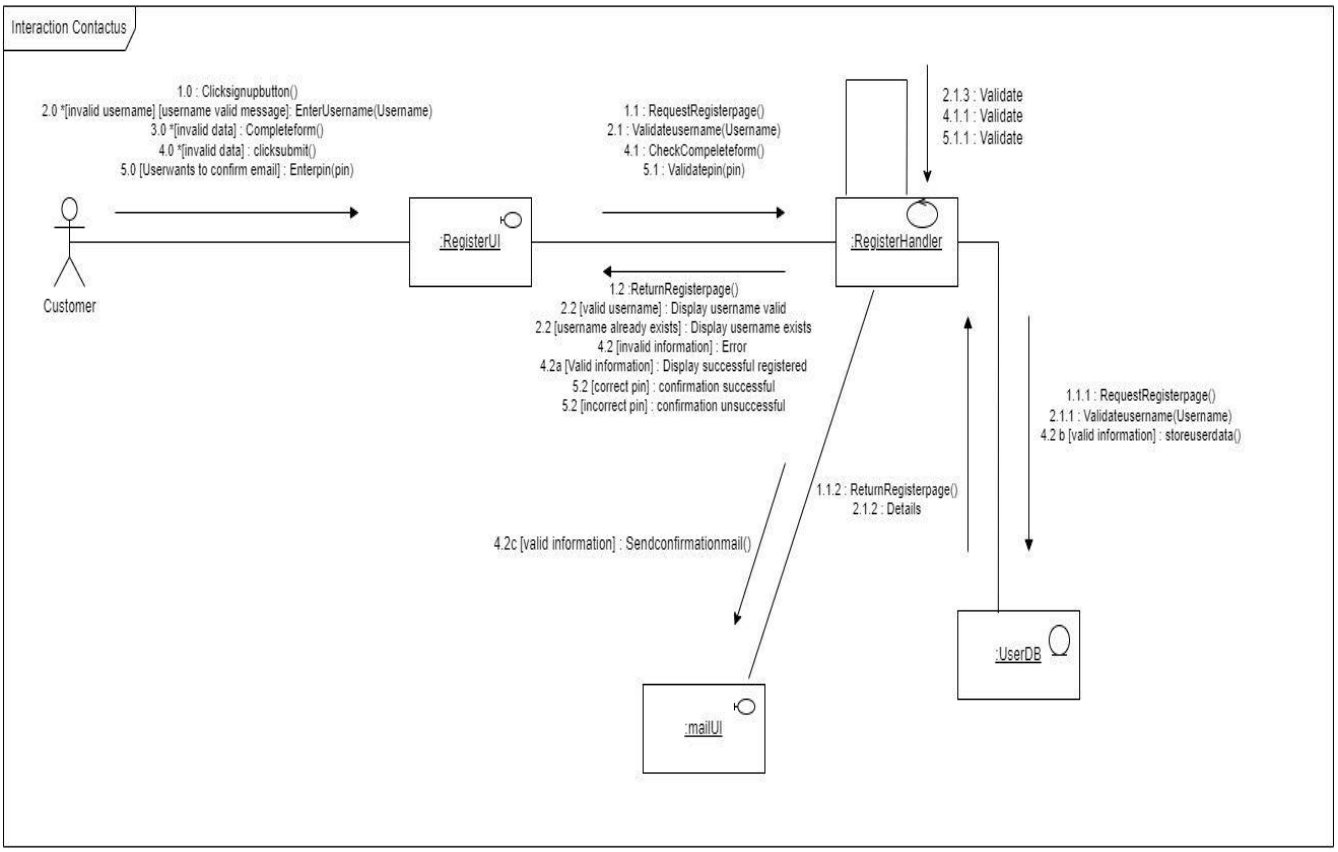
### Register




Sequence Diagram

## Communication Diagrams

Register




Communication Diagram

**Contributions**

| Registration No. | Name | Function |
|---|---|---|
| IT23256750 | W.M.V. Wijethunge | Create a new trip |
| IT23252936 | W.L.S.A. Wijesinha | Payment process |
| IT23243026 | R.M.U.K. Rathnayake | Driver update profile |
| IT23150966 | K.Y.T. Chamuditha | User register |